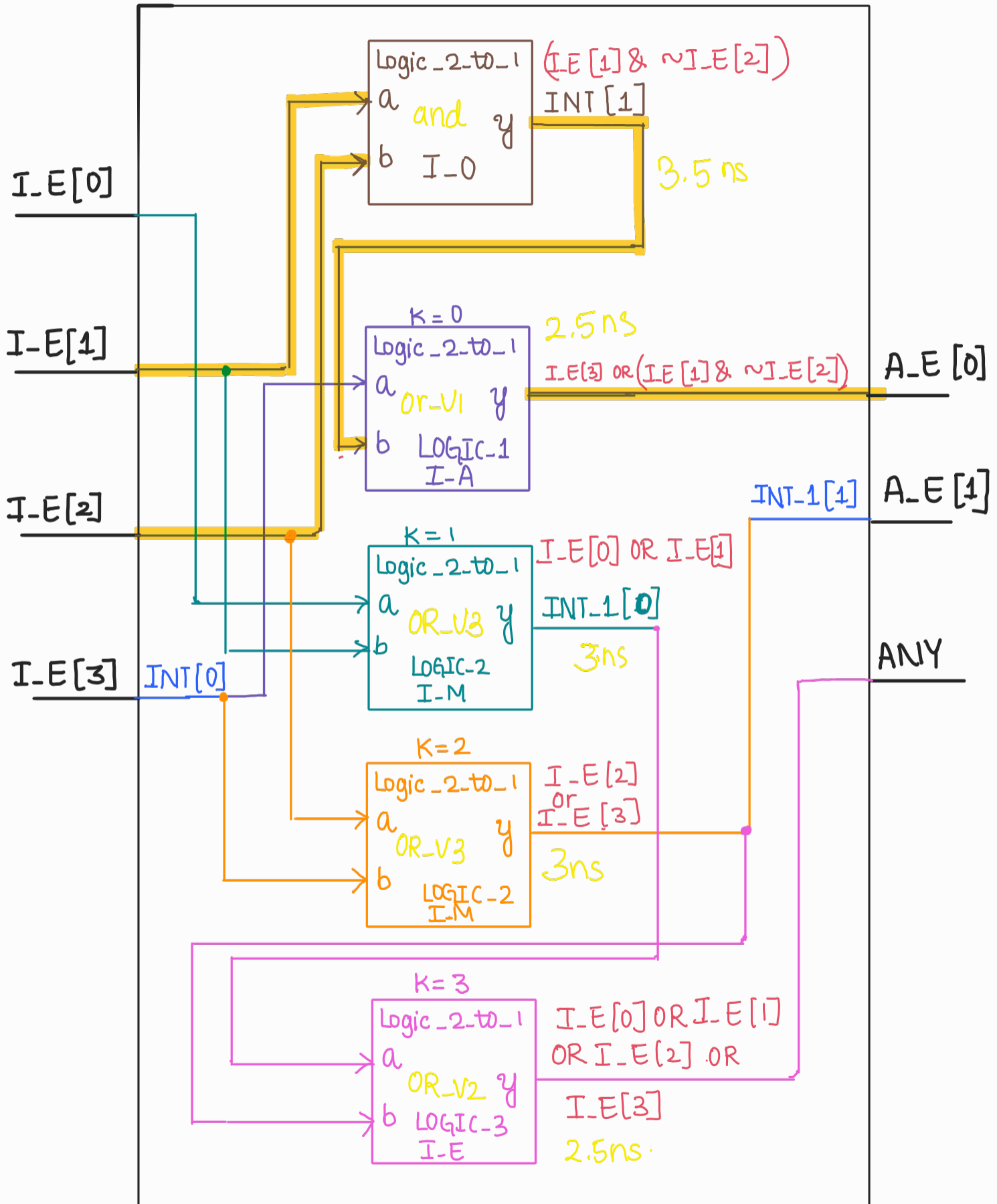


PREPARATION TASKS

Exercise 1

Input : I-E[0], I-E[1], I-E[2], I-E[3]
output : A-E[0], A-E[1], ANY
Signals : INT[0], INT[1], INT_1[0], INT_1[1]



ENCODER_4-2-2

Critical path.

Exercise 2

I-E[3]	I-E[2]	I-E[1]	I-E[0]	A-E[1]	A-E[0]	ANY	
0	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0
0	0	1	0	0	1	1	1
0	0	1	1	0	1	1	1
0	1	0	0	1	0	1	2
0	1	0	1	1	0	1	2
0	1	1	0	1	0	1	2
0	1	1	1	1	0	1	2
1	0	0	0	1	1	1	3
1	0	0	1	1	1	1	3
1	0	1	0	1	1	1	3
1	0	1	1	1	1	1	3
1	1	0	0	1	1	1	3
1	1	0	1	1	1	1	3
1	1	1	0	1	1	1	3
1	1	1	1	1	1	1	3

$$A-E[0] = I-E[3] \text{ OR } (I-E[1] \& \sim I-E[2])$$

$$A-E[1] = I-E[2] \text{ OR } I-E[3]$$

$$ANY = I-E[0] \text{ OR } I-E[1] \text{ OR } I-E[2] \text{ OR } I-E[3]$$

→

The design is a 4 to 2 bit Priority Encoder.

If $I-E[3] = 1$, then output : "11" irrespective of others

If $I-E[2] = 1$, then output : "10" irrespective of $I-E[1]$ & $I-E[0]$
When $I-E[3] = 0$

If $I-E[1] = 1$, then output : "01" irrespective of $I-E[0]$

If $I-E[0] = 1$, then output : "00" and $ANY = '1'$

If all inputs are 0, $A-E = 00$ & $ANY = 0$

Means input is active
& is applied.

Exercise 3

Checking all signals with highest delay
of respective architectures.

Paths :

$I-E[1], I-E[2] \rightarrow I_0 \rightarrow INT[1] \rightarrow I_A \rightarrow A-E[0] \quad 3.5 + 2.5 = 6ns$

$INT[1], I-E[3] \rightarrow I_A \rightarrow A-E[0] \quad 2.5 ns.$

$I-E[0], I-E[1] \rightarrow I-M(k=1) \rightarrow INT-1[0] \rightarrow I-E(k=3) \rightarrow ANY$
 $3ns + 2.5ns = 5.5ns$

$I-E[2], I-E[3] \rightarrow I-M(k=2) \rightarrow INT-1[1] \rightarrow I-E(k=3) \rightarrow ANY$
 $3ns + 2.5ns = 5.5ns$

$I-E[2], I-E[3] \rightarrow I-M(k=2) \rightarrow INT-1[1] \rightarrow A-E[1] \quad 3ns$

$I-E[1], I-E[2] \rightarrow I_0 \rightarrow INT[1] \rightarrow I_A \rightarrow A-E[0]$

is the critical path and the delay in this path is

$$3.5ns + 2.5ns = 6ns$$

(and-2-1-2n) + (OR-v1)

